

Basic Econometrics

Course Objectives

The purpose of this course is to provide the participants with the necessary understanding of econometric theory and concepts and to know what econometric methods to use in different circumstances and how to interpret and appraise the results of the empirical analysis.

Course Contents

1. Introduction

- Econometrics a separate discipline
- Methodology of Econometrics
- Expected Value and its properties
- Types of data

2. Two Variable Regression Model

- The method of Ordinary Least Square
- Assumptions of Classical Linear Regression Model
- Derivation of Least Square Estimates
- Variance of disturbance term
- Best linear unbiased property (BLUE) of an estimator
- Regression through origin
- Elasticity and slope

3. Two Variable Regression: Interval estimation & Hypothesis Testing

- Interval estimation: some basic ideas
- Confidence intervals for regression coefficients
- R^2 Goodness of fit
- Hypothesis testing: The test of significance approach
 - Testing the significance of regression coefficients
 - The t-test
- Regression analysis and analysis of variance (ANOVA)

4. Multiple Regression Analysis

- Introduction
- Meaning of partial regression coefficients
- Multiple regression model in deviation form
- OLS estimation
- Variance covariance matrix of β
- Properties of OLS vector β (BLUE property of β)
- Coefficient of determination R^2
- Hypothesis testing for partial regression coefficients
- Testing the overall significance of regression

5. Specification Errors

- Meaning of specification errors
- Left out variable and irrelevant variable
- Measurement errors in the dependent variable and explanatory variable
- Wrong functional form

- Consequences of specification errors
- Detection of specification bias

6. Regression on Dummy variables

- Meaning of dummy variables
- Examples of dummy variables
- Comparing regressions (Chow and dummy variable test) with examples

7. Multicollinearity

- Meaning and the nature of multicollinearity
- Perfect and imperfect multicollinearity
- Consequences of multicollinearity
- Detection of multicollinearity
- Remedial measures

8. Heteroscedasticity

- Meaning and the nature of heteroscedasticity
- Consequences of heteroscedasticity
- Detection of heteroscedasticity
- Remedial measures

9. Autocorrelation

- Meaning and the nature of autocorrelation
- Consequences of autocorrelation
- Detection of autocorrelation
- Remedial measures

Recommended Books

1. Gujarati, D. N, and Sangeetha (2017). *Basic Econometrics*. 6th Edition. McGraw-Hill Companies.
2. Brooks, C. (2008). *Introductory Econometrics for Finance*. Cambridge University Press.
3. Dougherty, C. (2007). *Introduction to Econometrics*. 3rd Edition. Oxford University Press.
4. Koutsoyiannis, A. (1992). *Theory of Econometrics*. Macmillan Education Ltd.
5. Patterson, K. (2000). *An Introduction to Applied Econometrics*. Palgrave Macmillan.

Note: 1. Duration of course will be 10 weeks.

2. Total 10 lectures of two hours duration will be delivered.

3. Class time will be from 7:00 pm to 9:00 pm, once a week.

4. Atleast 75% attendance is required to get a certificate.